Lie Groups, Lie Algebras, and Representations
Reading Notes and Exercises

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Contents

1 Matrix Lie Groups

1.1 Definitions

Definition 1.1. *The **general linear group** over the real numbers, denoted $GL(n,\mathbb{R})$, is the group of all $n \times n$ invertible matrices with real entries. The general linear group over the complex numbers, denoted $GL(n,\mathbb{C})$, is the group of all $n \times n$ invertible matrices with complex entries.*

Definition 1.2. *Let $M_n(\mathbb{C})$ denote the space of all $n \times n$ matrices with complex entries.*

Definition 1.3. *Let A_m be a sequence of complex matrices in $M_n(\mathbb{C})$. We say A_m **converges** to a matrix A if each entry of A_m converges to the corresponding entry of A as $m \to \infty$, i.e. if $(A_m)_{jk}$ converges to A_{jk} for all $1 \leq j, k \leq n$.*

Note that Definition 1.3 means we can identify $M_n(\mathbb{C})$ with $\mathbb{C}^{n^2}$, and use the standard notion of convergence in $\mathbb{C}^{n^2}$. All topological properties of a matrix Lie group G will be considered with respect to the topology G inherits as a subset of $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$.

We now consider subgroups of $GL(n,\mathbb{C})$, which are subsets $G \subset GL(n,\mathbb{C})$ such that $Id \in G$, and for all $A, B \in G$ we have $AB \in G$ and $A^{-1} \in G$.

Definition 1.4. *A **matrix Lie group** is a subgroup G of $GL(n,\mathbb{C})$ with the property: if A_m is any sequence of matrices in G, and A_m converges to some matrix A, then either A is in G or A is not invertible.*

This property means that G is a closed subset of $GL(n,\mathbb{C})$. So this definition is equivalent to saying a matrix Lie group is a closed subgroup of $GL(n,\mathbb{C})$. However, G is not necessarily closed in $M_n(\mathbb{C})$. For this to be true, we require a stronger property: if A_m is any sequence of matrices in G and A_m converges to some matrix A, then $A \in G$.

1.2 Examples

We introduce some important examples of (matrix) Lie groups, including the classical groups.

1.2.1 General and Special Linear Groups

The general linear groups $GL(n,\mathbb{R})$ and $GL(n,\mathbb{C})$, defined in 1.1, are matrix Lie groups.

The special linear groups $SL(n,\mathbb{R})$ and $SL(n,\mathbb{C})$ is the group of $n \times n$ invertible matrices (with real or complex entries) having determinant 1. They are matrix Lie groups.

1.2.2 Unitary and Orthogonal Groups

A $n \times n$ complex matrix A is **unitary** if its column vectors are orthonormal, that is,

$$\sum_{l=1}^{n} \bar{A}_{lj} A_{lk} = \delta_{jk}$$

where δ_{jk} is the Kronecker delta. This can be rewritten as

$$\sum_{l=1}^{n} (A^*)_{jl} A_{lk} = \delta_{jk}$$

where A^* is the adjoint of A, defined by $A^* = \bar{A}^t$. A is unitary if and only if $A^* = A^{-1}$.

The **unitary group** $U(n)$ is the collection of all unitary matrices. The **special unitary group** $SU(n)$ is the subgroup of $U(n)$ consisting of unitary matrices with determinant 1. $U(n)$ and $SU(n)$ are matrix Lie groups.

A $n \times n$ real matrix A is **orthogonal** if its column vectors are orthonormal. The equivalent conditions are the same as before, except that when A is real, $A^* = A^t$. A is orthogonal if and only if $A^t = A^{-1}$, if and only if A preserves the inner product on $\mathbb{R}^n$.

The **orthogonal group** $O(n)$ is the collection of all orthogonal matrices. The **special orthogonal group** $SO(n)$ is the set of $n \times n$ orthogonal matrices with determinant 1. Elements of $SO(n)$ are rotations, while the elements of $O(n)$ are either rotations or combinations of rotations and reflections.

Consider the bilinear form $(\cdot, \cdot)$ on $\mathbb{C}^n$ defined by

$$(x, y) = \sum_j x_j y_j$$

Note that this form is not an inner product, since it is symmetric, not conjugate-symmetric. The **complex orthogonal group** $O(n, \mathbb{C})$ is the set of all $n \times n$ complex matrices A that preserve this form. The group $SO(n, \mathbb{C})$ is the set of all matrices in $O(n, \mathbb{C})$ with determinant 1. $O(n, \mathbb{C})$ and $SO(n, \mathbb{C})$ are matrix Lie groups.

1.2.3 Generalized Orthogonal and Lorentz Groups

Consider $\mathbb{R}^{n+k}$, with n and k positive integers. Define a symmetric bilinear form $[\cdot, \cdot]_{n,k}$ on $\mathbb{R}^{n+k}$ by

$$[x, y]_{n,k} = x_1 y_1 + \cdots + x_n y_n - x_{n-1} y_{n-1} - \cdots - x_{n+k} y_{n+k}$$

The **generalized orthogonal group** $O(n, k)$ is the set of $(n+k) \times (n+k)$ real matrices A that preserve this form, i.e. $[Ax, Ay]_{n,k} = [x, y]_{n,k}$. It is a matrix Lie group. In particular, the **Lorentz group** is $O(3, 1)$. $SO(n, k)$ is the subgroup of $O(n, k)$ consisting of elements with determinant 1.

For any $A \in O(n, k)$, we have $\det A = \pm 1$.

1.2.4 Symplectic Groups

Define the skew-symmetric bilinear form ω on $\mathbb{R}^{2n}$ by

$$\omega(x, y) = \sum_{j=1}^{n} (x_j y_{n+j} - x_{n+j} y_j)$$

The **real symplectic group** $Sp(n, \mathbb{R})$ is the set of all $2n \times 2n$ real matrices A which preserve ω. It is a closed subgroup of $GL(2n, \mathbb{R})$, hence a matrix Lie group. $\det A = \pm 1$ for all $A \in Sp(n, \mathbb{R})$; in fact, $\det A = 1$.

Using the same formula, we can define a bilinear form ω on $\mathbb{C}^{2n}$. Over $\mathbb{C}$, we have the relation $\omega(z, w) = (z, \Omega w)$, where $(\cdot, \cdot)$ is the complex bilinear form from 1.2.2. The **complex symplectic group** $Sp(n, \mathbb{C})$ is the set of all $2n \times 2n$ complex matrices which preserve this form. We can show that for all $A \in Sp(n, \mathbb{C})$, $\det A = \pm 1$, and in fact, $\det A = 1$.

The **compact symplectic group** $Sp(n)$ is defined as

$$Sp(n) = Sp(n, \mathbb{C}) \cap U(2n)$$

That is to say, $Sp(n)$ is the group of $2n \times 2n$ matrices that preserve both the inner product and the bilinear form ω.

1.2.5 The Euclidean and Poincaré Groups

The **Euclidean group** $E(n)$ is the group of all transformations of $\mathbb{R}^n$ that can be expressed as a composition of a translation and an orthogonal linear transformation. Elements of $E(n)$ are written as pairs $\{x, R\}$ with $x \in \mathbb{R}^n$ and $R \in O(n)$, and $\{x, R\}$ acts on $\mathbb{R}^n$ by

$$\{x, R\}y = Ry + x$$

The product operation for $E(n)$ is

$$\{x_1, R_1\}\{x_2, R_2\} = \{x_1 + R_1 x_2, R_1 R_2\}$$

The inverse of an element of $E(n)$ is

$$\{x, R\}^{-1} = \{-R^{-1}x, R^{-1}\}$$

$E(n)$ is not a subgroup of $GL(n, \mathbb{R})$, but it is isomorphic to a closed subgroup of $GL(n+1, \mathbb{R})$.

The **Poincaré group** $P(n, 1)$ is the group of all transformations of $\mathbb{R}^{n+1}$ of the form

$$T = T_x A$$

with $x \in \mathbb{R}^{n+1}$ and $A \in O(n, 1)$.

1.2.6 The Heisenberg Group

The **Heisenberg group** H is the set of all 3×3 real matrices

$$A = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}$$

where $a, b, c \in \mathbb{R}$. H is a subgroup of $GL(3, \mathbb{R})$.

1.2.7 The Groups $\mathbb{R}^*$, $\mathbb{C}^*$, S^1, $\mathbb{R}$, and $\mathbb{R}^n$

These groups are not defined as groups of matrices, but can be thought of as such.

- The group $\mathbb{R}^*$ of non-zero real numbers under multiplication is isomorphic to $GL(1, \mathbb{R})$
- The group $\mathbb{C}^*$ of non-zero complex numbers under multiplication is isomorphic to $GL(1, \mathbb{C})$
- The group S^1 of complex numbers with absolute value 1 is isomorphic to $U(1)$
- The group $\mathbb{R}$ under addition is isomorphic to $GL(1, \mathbb{R})^+$, the 1×1 matrices with positive determinant
- The group $\mathbb{R}^n$ under vector addition is isomorphic to the group of diagonal real matrices with positive diagonal entries

1.2.8 The Compact Symplectic Group

We try to understand the structure of $Sp(n)$, and show that it can be understood as the "unitary group over the quaternions".

Define a conjugate-linear map $J : \mathbb{C}^{2n} \to \mathbb{C}^{2n}$ by

$$J(\alpha, \beta) = (-\bar{\beta}, \bar{\alpha})$$

where $\alpha, \beta \in \mathbb{C}^n$ and $(\alpha, \beta) \in \mathbb{C}^{2n}$. For all $z, w \in \mathbb{C}^{2n}$, we have

$$\omega(z, w) = \langle Jz, w \rangle$$

Furthermore, for all $z, w \in \mathbb{C}^{2n}$, we have

$$\langle Jz, w \rangle = -\langle Jw, z \rangle$$

and

$$J^2 = -I$$

Proposition 1.5. *If $U \in U(2n)$, then $U \in Sp(n)$ if and only if U commutes with J.*

Theorem 1.6. *$U \in Sp(n)$ if and only if there exists an orthonormal basis $u_1, \ldots, u_n, v_1, \ldots, v_n$ for $\mathbb{C}^{2n}$ such that the following properties hold:*

1. *$Ju_j = v_j$*
2. *For some real numbers $\theta_1, \ldots, \theta_n$, we have*

$$\begin{aligned} Uu_j &= e^{i\theta_j} u_j \\ Uv_j &= e^{-i\theta_j} v_j \end{aligned}$$

3. *We have*

$$\begin{aligned} \omega(u_j, u_k) &= \omega(v_j, v_k) = 0 \\ \omega(u_j, v_k) &= \delta_{jk} \end{aligned}$$

Lemma 1.7. *Suppose V is a complex subspace of $\mathbb{C}^{2n}$ that is invariant under the conjugate-linear map J. Then the orthogonal complement $V^\perp$ of V (with respect to the inner product $\langle \cdot, \cdot \rangle$) is also invariant under J. Furthermore, V and $V^\perp$ are orthogonal with respect to ω, i.e.*

$$\omega(z, w) = 0$$

for all $z \in V$ and $w \in V^\perp$.

1.3 Topological Properties

1.3.1 Compactness

Definition 1.8. *A matrix Lie group $G \subset GL(n, \mathbb{C})$ is **compact** if it is compact in the usual topological sense as a subset of $M_n(\mathbb{C}) \cong \mathbb{R}^{2n^2}$.*

By the Heine-Borel theorem, a matrix Lie group G is compact if and only if it is closed and bounded.

$O(n)$, $SO(n)$, $U(n)$, $SU(n)$, and $Sp(n)$ are compact.

1.3.2 Connectedness

Definition 1.9. *A matrix Lie group G is **connected** if for all $A, B \in G$, there exists a continuous path $A(t)$, $a \leq t \leq b$, lying in G with $A(a) = A$ and $A(b) = B$. For any matrix Lie group G, the **identity component** of G, denoted G_0, is the set of $A \in G$ for which there exists a continuous path $A(t)$, $a \leq t \leq b$, lying in G with $A(a) = I$ and $A(b) = A$.*

This definition is actually called **path connected** in topology. A matrix Lie group is connected if and only if it is path-connected.

To show that a matrix Lie group G is connected, it suffices to show that each $A \in G$ can be connected to the identity by a continuous path lying in G.

Proposition 1.10. *If G is a matrix Lie group, the identity component G_0 of G is a normal subgroup of G.*

G_0 is closed, and hence a matrix Lie group.

Proposition 1.11. *$GL(n,\mathbb{C})$ is connected for all $n \geq 1$.*

Proposition 1.12. *$SL(n,\mathbb{C})$ is connected for all $n \geq 1$.*

Proposition 1.13. *$U(n)$, $SU(n)$, and $SO(n)$ are connected for all $n \geq 1$.*

1.3.3 The Topology of $SO(3)$

Definition 1.14. *The **real projective space** of dimension n, denoted $\mathbb{R}P^n$, is the set of lines through the origin in $\mathbb{R}^{n+1}$. Since each line through the origin intersects the unit sphere exactly twice, we may think of $\mathbb{R}P^n$ as the unit sphere S^n with "antipodal" points u and $-u$ identified.*

Points in $\mathbb{R}P^n$ can be thought of as pairs $\{u, -u\}$ with $u \in S^n$. There is a natural map $\pi : S^n \to \mathbb{R}P^n$

$$\pi(u) = \{u, -u\}$$

A distance function can be defined on $\mathbb{R}P^n$ by

$$d(\{u, -u\}, \{v, -v\}) = \min(d(u, v), d(u, -v))$$

The space $\mathbb{R}P^n$ is the upper hemisphere in S^n with antipodal points on the equator identified. $\mathbb{R}P^n$ is the closed unit ball $B^n \subset \mathbb{R}^n$, with antipodal points on the boundary of B^n identified.

Proposition 1.15. *There is a continuous bijection between $SO(3)$ and $\mathbb{R}P^n$.*

$\mathbb{R}P^n$ is not simply connected, and hence $SO(3)$ is not simply connected.

1.4 Homomorphisms

Definition 1.16. *Let G and H be matrix Lie groups. A map Φ from G to H is called a **Lie group homomorphism** if*

1. *Φ is a group homomorphism, and*
2. *Φ is continuous.*

*If, in addition, Φ is one-to-one and onto and the inverse map Φ^{-1} is continuous, then Φ is called a **Lie group isomorphism**.*

If G and H are matrix Lie groups, and there exists a Lie group isomorphism from G to H, then G and H are isomorphic, $G \cong H$.

The relationship between $SU(2)$ and $SO(3)$: they are not isomorphic, but there exists a Lie group homomorphism $\Phi : SU(2) \to SO(3)$ that is two-to-one and onto.

Proposition 1.17. *The map $U \Rightarrow \Phi_U$ is a 2-1 and onto map of $SU(2)$ to $SO(3)$, with kernel equal to $\{I, -I\}$.*

$SU(2)$ is homeomorphic to S^3. The proposition shows that $SO(3)$ is homeomorphic to $\mathbb{R}P^3$, that is, S^3 with antipodal points identified.

1.5 Lie Groups

The text first gives a very brief discussion of manifolds and smooth manifolds.

Definition 1.18. *A* ***Lie group*** *is a smooth manifold G which is also a group, and such that the group product $G \times G \to G$ and the inverse map $G \to G$ are smooth.*

Every matrix Lie group is a Lie group, but the reverse is not true. However, we focus on matrix Lie groups.

Usually, we call a map Φ between two Lie groups a Lie group homomorphism if Φ is a group homomorphism and Φ is smooth.

Every matrix Lie group G is a manifold, so G must be locally path connected, so G is connected if and only if it is path connected.

1.6 Exercises

Exercise 1.7 Using Theorem 1.6, show that $Sp(n)$ is connected and that every element of $Sp(n)$ has determinant 1.

Proof. Let $U \in Sp(n)$. Note that the matrix U represents a linear transformation $L_u : \mathbb{C}^{2n} \to \mathbb{C}^{2n}$. By Theorem 1.6, there exists an orthonormal basis $u_1, \ldots, u_n, v_1, \ldots, v_n$ for $\mathbb{C}^{2n}$ such that

1. $Ju_j = v_j$
2. for some $\theta_1, \ldots, \theta_n \in \mathbb{R}$,
$$Uu_j = e^{i\theta_j} u_j$$
$$Uv_j = e^{-i\theta_j} v_j$$
3. and we have
$$\omega(u_j, u_k) = \omega(v_j, v_k) = 0$$
$$\omega(u_j, v_k) = \delta_{jk}$$

The second point is the most important. Notice that it implies $e^{i\theta_j}$ are eigenvalues of U with eigenvectors u_j, and $e^{-i\theta_j}$ are eigenvalues of U with eigenvectors v_j.

Denote $B = \{u_1, \ldots, u_n, v_1, \ldots, v_n\}$. B is a basis for $\mathbb{C}^{2n}$ consisting entirely of eigenvectors of U, so U is diagonalizable. Thus there exists an invertible matrix $Q \in M_{2n}(\mathbb{C})$ such that

$$QUQ^{-1} = diag(e^{i\theta_1}, \ldots, e^{i\theta_n}, e^{-i\theta_1}, \ldots, e^{-i\theta_n})$$

and so

$$U = Q^{-1} diag(e^{i\theta_1}, \ldots, e^{i\theta_n}, e^{-i\theta_1}, \ldots, e^{-i\theta_n})Q$$

Now, we make a slight modification and define

$$U(t) = Q^{-1} diag(e^{i(1-t)\theta_1}, \ldots, e^{i(1-t)\theta_n}, e^{-i(1-t)\theta_1}, \ldots, e^{-i(1-t)\theta_n})Q$$

for $0 \le t \le 1$. $U(t)$ is continuous, $U(0) = U$, and $U(1) = Q^{-1}IQ = I$. By applying the converse of Theorem 1.6 to $U(t)$, it is clear that $U(t) \in Sp(n)$ for all $0 \le t \le 1$.

Hence $Sp(n)$ is connected.

For any $U \in Sp(n)$,

$$\begin{aligned}
\det(U) &= \det(Q^{-1} diag(e^{i\theta_1}, \ldots, e^{i\theta_n}, e^{-i\theta_1}, \ldots, e^{-i\theta_n})Q) \\
&= \det(Q^{-1}) \det(diag(e^{i\theta_1}, \ldots, e^{i\theta_n}, e^{-i\theta_1}, \ldots, e^{-i\theta_n})) \det(Q) \\
&= \frac{1}{\det(Q)} e^{i\theta_1} \cdots e^{i\theta_n} e^{-i\theta_1} \cdots e^{-i\theta_n} \det(Q) \\
&= 1
\end{aligned}$$

□

Exercise 1.9 Suppose a is an irrational real number. Show that the set E_a of numbers of the form $e^{2\pi ina}$, $n \in \mathbb{Z}$, is dense in the unit circle S^1.

Proof. We follow the given hint, and divide S^1 into N equally sized bins of length $\frac{2\pi}{N}$. E_a has an infinite number of elements, and N may be large but finite. There must be at least one bin that has an infinite number of elements of E_a (otherwise, if every bin has finite number of elements and N is finite, then E_a would be finite).

Now, choose a bin with infinitely many elements of E_a. In this bin, it is possible to choose two distinct elements of E_a of the form $e^{2\pi ina}$ and $e^{2\pi ima}$, with $n, m \in \mathbb{R}$, $n \neq m$.

We use the fact that E_a is a subgroup of S^1. $e^{2\pi ina}, e^{2\pi ima} \in E_a$, so the inverse $e^{-2\pi ima} \in E_a$, so $e^{2\pi i(n-m)a} \in E_a$, and so $(e^{2\pi i(n-m)a})^k = e^{2k\pi i(n-m)a} \in E_a$ for any $k \in \mathbb{Z}$.

By taking N to be large, we can force $e^{2\pi ina}$ and $e^{2\pi ima}$ to be arbitrarily close to each other on S^1. Then $e^{2k\pi i(n-m)a}$ is a rotation by an arbitrarily small angle. Putting it to the power of k allows us to approximate any point on S^1 by an element of E_a.

More formally, let $e^{i\theta} \in S^1$ for any $\theta \in \mathbb{R}$.

$$
\begin{aligned}
|e^{i\theta} - e^{2k\pi i(n-m)a}| &= \left| e^{2k\pi i(n-m)a} \left(\frac{e^{i\theta}}{e^{2k\pi i(n-m)a}} - 1 \right) \right| \\
&= \left| \frac{e^{i\theta}}{e^{2k\pi i(n-m)a}} - 1 \right| \\
&= |e^{i(\theta - 2k\pi(n-m)a)} - 1| \\
&= |e^{i(\theta - 2k\pi(n-m)a)} - e^0|
\end{aligned}
$$

By continuity, $\forall \varepsilon > 0$, $\exists \delta > 0$ such that $|\theta - 2k\pi(n-m)| < \delta \Rightarrow |e^{i(\theta - 2k\pi(n-m)a)} - e^0| < \varepsilon$. So $\forall \varepsilon > 0$, we can choose N large, and a suitable k such that

$$|e^{i\theta} - e^{2k\pi i(n-m)a}| < \varepsilon$$

Hence E_a is dense in S^1. □

2 The Matrix Exponential

2.1 The Exponential of a Matrix

If X is a $n \times n$ matrix, we define the **exponential** of X, denoted e^X, by the power series

$$e^X = \sum_{m=0}^{\infty} \frac{X^m}{m!} \tag{2.1}$$

where X^0 is the identity matrix I and X^m is the repeated matrix product of X with itself.

Proposition 2.1. *The series 2.1 converges for all $X \in M_n(\mathbb{C})$ and e^X is a continuous function of X.*

Definition 2.2. *For any $X \in M_n(\mathbb{C})$, we define the **Hilbert-Schmidt** norm of X,*

$$||X|| = \left(\sum_{j,k=1}^{n} |X_{jk}|^2 \right)^{1/2}$$

The Hilbert-Schmidt norm may be computed in a basis-independent way as

$$||X|| = (tr(X^* X))^{1/2}$$

This norm satisfies

$$||X + Y|| \leq ||X|| + ||Y||$$
$$||XY|| \leq ||X|| ||Y||$$

for all $X, Y \in M_n(\mathbb{C})$.

Proposition 2.3. *Let X and Y be arbitrary $n \times n$ matrices. Then we have the following:*

1. $e^0 = I$
2. $(e^X)^* = e^{X^*}$
3. e^X *is invertible and* $(e^X)^{-1} = e^{-X}$
4. $e^{(\alpha+\beta)X} = e^{\alpha X} e^{\beta X}$ *for all* α *and* β *in* $\mathbb{C}$
5. *If* $XY = YX$*, then* $E^{X+Y} = e^X e^Y = e^Y e^X$
6. *If* $C \in GL(N, \mathbb{C})$*, then* $e^{CXC^{-1}} = Ce^X C^{-1}$

Proposition 2.4. *Let X be a $n \times n$ complex matrix. Then e^{tX} is a smooth curve in $M_n(\mathbb{C})$ and*

$$\frac{d}{dt} e^{tX} = X e^{tX} = e^{tX} X$$

In particular,

$$\left. \frac{d}{dt} e^{tX} \right|_{t=0} = X$$

2.2 Computing the Exponential

The book explains some methods for computing the matrix exponential. If X is diagonalizable, then $X = CDC^{-1}$, and $e^X = Ce^D C^{-1}$. If X is nilpotent, ($X^k = 0$ for some k), then the power series of e^X terminates. Finally, every matrix X can be written uniquely as $X = S + N$ where S is diagonalizable, N is nilpotent, and $SN = NS$. Then $e^X = e^{S+N} = e^S e^N$.

2.3 The Matrix Logarithm

We want to define a matrix logarithm, which should be an inverse function to the matrix exponential. The simplest way is by a power series. Recall that in the case of complex numbers,

Lemma 2.5. *The function*

$$\log z = \sum_{m=1}^{\infty} (-1)^{m+1} \frac{(z-1)^m}{m}$$

is defined and holomorphic in a circle of radius 1 about $z = 1$. For all z with $|z-1| < 1$,

$$e^{\log z} = z$$

For all u with $|u| < \log 2$, we have $|e^u - 1| < 1$ and

$$\log e^u = u$$

Definition 2.6. *For an $n \times n$ matrix A, define $\log A$ by*

$$\log A = \sum_{m=1}^{\infty} (-1)^{m+1} \frac{(A-I)^m}{m}$$

whenever the series converges.

Theorem 2.7. *The function*

$$\log A = \sum_{m=1}^{\infty} (-1)^{m+1} \frac{(A-I)^m}{m}$$

is defined and continuous on the set of all $n \times n$ complex matrices A with $||A - I|| < 1$. For all $A \in M_n(\mathbb{C})$ with $||A - I|| < 1$,

$$e^{\log A} = A$$

For all $X \in M_n(\mathbb{C})$ with $||X|| < \log 2$, $||e^X - I|| < 1$ and

$$\log e^X = X$$

Notice from the last sentence that it is not true that $\log e^X = X$ whenever the series for the logarithm converges.

Proposition 2.8. *There exists a constant c such that for all $n \times n$ matrices B with $||B|| < 1/2$, we have*

$$|| \log(I + B) - B|| \leq c||B||^2$$

More concisely, the proposition can be stated as

$$\log(I + B) = B + O(||B||^2)$$

Theorem 2.9. *Every invertible $n \times n$ matrix can be expressed as e^X for some $X \in M_n(\mathbb{C})$.*

2.4 Further Properties of the Exponential

Theorem 2.10 (Lie Product Formula). *For all $X, Y \in M_n(\mathbb{C})$, we have*

$$e^{X+Y} = \lim_{m\to\infty} \left(e^{\frac{X}{m}} e^{\frac{Y}{m}}\right)^m$$

Theorem 2.11. *For any $X \in M_n(\mathbb{C})$, we have*

$$\det(e^X) = e^{tr(X)}$$

Definition 2.12. *A function $A : \mathbb{R} \to GL(n, \mathbb{C})$ is called a **one-parameter subgroup** of $GL(n, \mathbb{C})$ if*

1. *A is continuous,*
2. *$A(0) = I$,*
3. *$A(t + s) = A(t)A(s)$ for all $t, s \in \mathbb{R}$.*

Theorem 2.13 (One-Parameter Subgroups). *If $A(\cdot)$ is a one-parameter subgroup of $GL(n, \mathbb{C})$, there exists a unique $n \times n$ complex matrix X such that*

$$A(t) = e^{tX}$$

Lemma 2.14. *Fix some ε with $\varepsilon < \log 2$. Let $B_{\varepsilon/2}$ be the ball of radius $\varepsilon/2$ around the origin in $M_n(\mathbb{C})$, and let $U = \exp(B_{\varepsilon/2})$. Then every $B \in U$ has a unique square root C in U, given by $C = \exp(\frac{1}{2} \log B)$.*

Proposition 2.15. *The exponential map is an infinitely differentiable map of $M_n(\mathbb{C})$ into $M_n(\mathbb{C})$.*

2.5 The Polar Decomposition

The polar decomposition for a nonzero complex number z states that z can be written uniquely as $z = up$, where $|u| = 1$ and p is real and positive. Since p is real and positive, it can be written as $p = e^x$ for a unique real number x. This gives the polar decomposition for z,

$$z = ue^x$$

with $x \in \mathbb{R}$ and $|u| = 1$.

We want to establish a similar polar decomposition for $GL(n, \mathbb{C})$, then for various subgroups. If P is a self-adjoint $n \times n$ matrix ($P^* = P$), we say that P is **positive** if $\langle v, Pv\rangle > 0$ for all nonzero $v \in \mathbb{C}^n$. A self-adjoint matrix P is positive if and only if all the eigenvalues of P are positive.

Theorem 2.16. *Polar decomposition for $GL(n, \mathbb{C})$.*

1. *Every $A \in GL(n, \mathbb{C})$ can be written uniquely in the form*

   $$A = UP$$

   where U is unitary and P is self-adjoint and positive.

2. *Every self-adjoint positive matrix P can be written uniquely in the form*

   $$P = e^X$$

   with X self-adjoint. Conversely, if X is self-adjoint, then e^X is self-adjoint and positive.

3. *If we decompose each $A \in GL(n, \mathbb{C})$ (uniquely) as*

$$A = Ue^X$$

with U unitary and X self-adjoint, then U and X depend continuously on A.

Lemma 2.17. *If Q is a self-adjoint, positive matrix, then Q has a unique self-adjoint, positive square root.*

Proof. Recall that a matrix Q is self-adjoint (Hermitian) if $Q^* = Q$, and unitary if $Q^* = Q^{-1}$. From Appendix A, we know that a matrix Q self-adjoint $\Rightarrow$ normal $\Rightarrow$ diagonalizable, and it is possible to find an orthonormal basis of eigenvectors for Q. Here, we can write $Q = UDU^{-1}$ where U can be taken to be unitary.

Let Q be a self-adjoint, positive matrix, then Q can be written as

$$Q = U \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} U^{-1}$$

with U unitary. The λ_j are eigenvalues corresponding to the orthonormal basis of eigenvectors of Q. Since Q is self-adjoint and positive, $\lambda_j > 0$ for all j.

Thus, we can construct a square root of Q as

$$Q^{1/2} = U \begin{pmatrix} \lambda_1^{1/2} & & \\ & \ddots & \\ & & \lambda_n^{1/2} \end{pmatrix} U^{-1}$$

$Q^{1/2}$ is still self-adjoint and positive. This proves the existence of the square root.

Consider P a self-adjoint, positive matrix with eigenvalues λ_j and eigenvectors v_j, and so

$$Pv_j = \lambda_j v_j$$

Notice that the eigenspaces of P^2 are the same as the eigenspaces of P, with the eigenvalues of P^2 equal to the squares of the eigenvalues of P. This is because

$$P^2 v_j = P(Pv_j) = P(\lambda_j v_j) = \lambda_j(\lambda_j v_j) = \lambda_j^2 v_j$$

The mapping $x \mapsto x^2$ is injective for $x \in R^{\geq 0}$. So eigenspaces with distinct eigenvalues remain with distinct eigenvalues after squaring.

If a self-adjoint positive Q has a self-adjoint positive square root P, then the eigenspaces of P must be the same as the eigenspaces of Q, and the eigenvalues of P must be the positive square roots of the eigenvalues of Q. Thus, P is uniquely determined by Q, proving uniqueness. □

Proof. of Theorem 2.16.

1. Note that if $A = UP$, then $A^*A = P^*U^*UP = PU^{-1}UP = P^2$. For any matrix A, the matrix A^*A is self-adjoint, since $(A^*A)^* = A^*A^{**} = A^*A$. If, in addition, A is invertible, then for all nonzero $v \in \mathbb{C}^n$, we have

$$\langle v, A^*Av \rangle = \langle Av, Av \rangle > 0$$

where the last inequality is due to the definition of the inner product and v being nonzero. So A^*A is positive. For all invertible A, let us define P by

$$P = (A^*A)^{1/2}$$

where the square root is the unique self-adjoint positive square root from Lemma 2.17. Then, we define

$$U = AP^{-1} = A[(A^*A)^{1/2}]^{-1}$$

P is self-adjoint and positive by construction, and $A = UP$ by the definition of U, so it remains only to check that U is unitary. To do this, we check

$$U^*U = [(A^*A)^{1/2}]^{-1}A^*A[(A^*A)^{1/2}]^{-1}$$

noting that the inverse of a self-adjoint positive matrix is self-adjoint. Since A^*A is the square of $(A^*A)^{1/2}$, we have

$$U^*U = [(A^*A)^{1/2}]^{-1}(A^*A)^{1/2}(A^*A)^{1/2}[(A^*A)^{1/2}]^{-1} = I$$

showing that U is unitary. This proves existence.

We already showed that if $A = UP$, then $A^*A = P^2$ where A^*A is self-adjoint positive. The uniqueness of P follows from Lemma 2.17. The uniqueness of U then follows from the definition $U = AP^{-1}$.

2. The existence and uniqueness of the decomposition is proven in the same way as in Lemma 2.17. Instead of the square root function, we use the logarithm function, which is a bijection between $(0, \infty)$ and $\mathbb{R}$.

   Let P be a self-adjoint matrix, so it is normal, so it is diagonalizable, and so P can be written as

$$P = U \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} U^{-1}$$

   where U can be taken to be unitary, and λ_j is an orthonormal basis of P. Since P is positive, $\lambda_j > 0$ for all j. Next, we define

$$\log P = U \begin{pmatrix} \log \lambda_1 & & \\ & \ddots & \\ & & \log \lambda_n \end{pmatrix} U^{-1} =: X$$

   Then,

$$P = \exp\left(U \begin{pmatrix} \log \lambda_1 & & \\ & \ddots & \\ & & \log \lambda_n \end{pmatrix} U^{-1} \right) = e^X$$

   We can check that X is self-adjoint

$$\begin{aligned} X^* &= (U^{-1})^* diag(\log \lambda_1, \dots, \log \lambda_n)^* U^* \\ &= (U^*)^{-1} diag(\log \lambda_1, \dots, \log \lambda_n)^* U^* \\ &= U diag(\log \lambda_1, \dots, \log \lambda_n) U^{-1} \\ &= X \end{aligned}$$

   So P can be written as e^X where X is self-adjoint, which proves existence.

   Moreover, the mapping $P \mapsto \log P$ is a bijection, so every P corresponds to a unique X such that $P = e^X$, which proves uniqueness.

   For a self-adjoint matrix X, we know from before that we can write

$$X = U \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} U^{-1}$$

with U unitary, and λ_j an orthonormal basis of X. Define

$$e^X = U \begin{pmatrix} e_1^\lambda & & \\ & \ddots & \\ & & e_n^\lambda \end{pmatrix} U^{-1}$$

e^X is self-adjoint because

$$\begin{aligned}(e^X)^* &= (U^{-1})^* diag(e_1^\lambda, \dots, e_n^\lambda)^* U^* \\ &= (U^*)^{-1} diag(e_1^\lambda, \dots, e_n^\lambda)^* U^* \\ &= U diag(e_1^\lambda, \dots, e_n^\lambda) U^{-1} \\ &= e^X\end{aligned}$$

Furthermore, e^X is positive because

$$\begin{aligned}\langle v, e^X v\rangle &= \langle v, U diag(e_1^\lambda, \dots, e_n^\lambda) U^{-1} v\rangle \\ &= \langle Uw, U diag(e_1^\lambda, \dots, e_n^\lambda) w\rangle \\ &= \langle w, U^* U diag(e_1^\lambda, \dots, e_n^\lambda) w\rangle \\ &= \langle w, diag(e_1^\lambda, \dots, e_n^\lambda) w\rangle \\ &= w^t diag(e_1^\lambda, \dots, e_n^\lambda) w \\ &= \begin{pmatrix} w_1 & \dots & w_n \end{pmatrix} \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} \\ &= \sum_{j=1}^{n} e^{\lambda_j} w_j^2 > 0\end{aligned}$$

Hence for any self-adjoint X self-adjoint, e^X is self-adjoint and positive.

X is referred to as the "logarithm" of P.

3. We first show that the logarithm X of a self-adjoint positive matrix P depends continuously on P. First, we consider the case where the eigenvalues of P are between 0 and 2. The proof of Theorem 2.7 establishes that if A is diagonalizable and all eigenvalues λ of A satisfy $|\lambda - 1| < 1$, then $\log A$ is defined and $e^{\log A} = A$. Similarly if X is diagonalizable and all eigenvalues λ of X satisfy $|\lambda| < \log 2$, then $\log e^X$ is defined and $\log e^X = X$. So the power series for $\log P$ converges to X.

$$\log P = \sum_{m=1}^{\infty} (-1)^{m+1} \frac{(P-I)^m}{m}$$

is defined for $||P - I|| < 1$. $(P - I)^m$ is a continuous function of P, so the partial sum is continuous. Note that

$$\left|(-1)^{m+1} \frac{(P-I)^m}{m}\right| = \frac{||(P-I)^m||}{m} \leq \frac{||P-I||^m}{m} \leq \frac{R^m}{m}, \quad R < 1$$

and that

$$\sum_{m=0}^{\infty} \frac{R^m}{m}$$

converges since

$$\lim_{m\to\infty} \left|\frac{R^{m+1}}{m+1} \frac{m}{R^m}\right| = \lim_{m\to\infty} R \frac{m}{m+1} = R < 1$$

By the Weierstrass M-test, the series converges uniformly on all sets of the form $\{||P - I|| \leq R\}$, for $R < 1$. So $\log P$ is continuous for all $P \in M_n(\mathbb{C})$ such that $||P - I|| < 1$.

Next, we consider the general case. Fix some self-adjoint positive matrix P_0. Fix a small neighbourhood $\mathcal{U}$ of P_0, choose $a > 0$ large. For $P \in \mathcal{U}$, write

$$P = e^a(e^{-a}P) = e^aI(e^{-a}P) = e^{aI}(e^{-a}P)$$

It is easy to see that e^{aI} and $e^{-a}P$ are commutative, and so

$$\log P = \log(e^{aI}(e^{-a}P)) = aI + \log(e^{-a}P)$$

If a is large enough, then for any $P \in \mathcal{U}$, the positive eigenvalues of $e^{-a}P$ all become less than 2, so the series for $\log(e^{-a}P)$ converge and depends continuously on P, thus X depends continuously on P.

By writing

$$P^{1/2} = \exp(\log(P)/2)$$

we see that the square root operation on P is also continuous. The mapping $A \mapsto A^*A$ is continuous, and the mapping $A^*A \mapsto (A^*A)^{1/2}$ is continuous. Since $P = (A^*A)^{1/2}$, P is continuous on A, and so X is continuous on A. Since $U = AP^{-1}$, U is continuous on A.

□

Proposition 2.18. *Polar decomposition for various subgroups of $GL(n, \mathbb{C})$.*

1. *Every $A \in GL(n, \mathbb{R})$ can be written uniquely as*

$$A = Re^X$$

where $R \in O(n)$ and X is real and symmetric.

2. *Every $A \in SL(n, \mathbb{C})$ can be written uniquely as*

$$A = Ue^X$$

where $U \in SU(n)$ and X is self-adjoint with trace zero.

3. *Every $A \in SL(n, \mathbb{R})$ can be written uniquely as*

$$A = Re^X$$

where $R \in SO(n)$ and X is real and symmetric with trace zero.

2.6 Exercises

Exercise 2.6 Show that every 2×2 matrix X with $tr(X) = 0$ satisfies

$$X^2 = -\det(X)I$$

If X is 2×2 with trace zero, show by direct calculation using the power series for the exponential that

$$e^X = \cos(\sqrt{\det X})I + \frac{\sin \sqrt{\det X}}{\sqrt{\det X}}X$$

where $\sqrt{\det X}$ is either of the two (possibly complex) square roots of $\det X$. Use this to give an alternative computation of the exponential e^{X_1} in Example 2.5.

Proof. Let X be a 2×2 matrix with $tr(X) = 0$, taking the form

$$X = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}$$

The determinant is

$$\det(X) = -a^2 - bc$$

Hence, we have

$$X^2 = \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \begin{pmatrix} a & b \\ c & -a \end{pmatrix} = \begin{pmatrix} a^2 + bc & 0 \\ 0 & a^2 + bc \end{pmatrix} = (a^2 + bc)I = -\det(X)I$$

which proves the required statement.

The power series of the matrix exponential is

$$e^X = \sum_{n=0}^{\infty} \frac{X^n}{n!} = I + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \frac{X^4}{4!} + \cdots$$

Using the result we just proved, $X^2 = -\det(X)I$, we can generalize and say that

$$X^{2n} = (-1)^n \det(X)^n I$$
$$X^{2n+1} = X^{2n}X = (-1)^n \det(X)^n X$$

The power series of the matrix exponential is absolutely convergent, so we can rearrange terms without worrying about the series converging to some other limit (source).

$$\begin{aligned}
e^X &= I + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \frac{X^4}{4!} + \cdots \\
&= (I + \frac{X^2}{2!} + \frac{X^4}{4!} + \cdots) + (X + \frac{X^3}{3!} + \frac{X^5}{5!} + \cdots) \\
&= \left(\sum_{n=0}^{\infty} \frac{(-1)^n \det(X)^n}{(2n)!} \right) I + \left(\sum_{n=0}^{\infty} \frac{(-1)^n \det(X)^n}{(2n+1)!} \right) X \\
&= \left(\sum_{n=0}^{\infty} \frac{(-1)^n \sqrt{\det X}^{2n}}{(2n)!} \right) I + \left(\sum_{n=0}^{\infty} \frac{(-1)^n \sqrt{\det X}^{2n+1}}{\sqrt{\det X}(2n+1)!} \right) X \\
&= \cos(\sqrt{\det X})I + \frac{\sin \sqrt{\det X}}{\sqrt{\det X}} X
\end{aligned}$$

where the last equality is from the power series of cos and sin. This proves the desired identity.

In Example 2.5, the matrix X_1 is

$$X_1 = \begin{pmatrix} 0 & -a \\ a & 0 \end{pmatrix}$$

which is a 2×2 matrix with trace zero. Its determinant is $\det X_1 = a^2$. We apply the above formula to compute its matrix exponential.

$$\begin{aligned}
e^{X_1} &= \cos(\sqrt{a^2})I + \frac{\sin \sqrt{a^2}}{\sqrt{a^2}} X_1 \\
&= \begin{pmatrix} \cos \sqrt{a^2} & 0 \\ 0 & \cos \sqrt{a^2} \end{pmatrix} + \begin{pmatrix} 0 & -a \frac{\sin \sqrt{a^2}}{\sqrt{a^2}} \\ a \frac{\sin \sqrt{a^2}}{\sqrt{a^2}} & 0 \end{pmatrix} \\
&= \begin{pmatrix} \cos a & -\sin a \\ \sin a & \cos a \end{pmatrix}
\end{aligned}$$

This is the same result as in Example 2.5. Note that $\sqrt{\det X_1} = \sqrt{a^2}$ can be taken to be either $+a$ or $-a$. The final result is the same. $\square$

3 Lie Algebras

3.1 Definitions and First Examples

Definition 3.1. *A **finite-dimensional real or complex Lie algebra** is a finite-dimensional real or complex vector space* $\mathfrak{g}$*, together with a map* $[\cdot,\cdot]$ *from* $\mathfrak{g}\times\mathfrak{g}$ *into* $\mathfrak{g}$*, with the following properties:*

1. $[\cdot,\cdot]$ *is bilinear.*
2. $[\cdot,\cdot]$ *is skew symmetric:* $[X,Y]=-[Y,X]$ *for all* $X,Y\in\mathfrak{g}$*.*
3. *The Jacobi identity holds:*

$$[X,[Y,Z]]+[Y,[Z,X]]+[Z,[X,Y]]=0$$

for all $X,Y,Z\in\mathfrak{g}$*.*

Two elements X and Y of $\mathfrak{g}$ commute if $[X,Y]=0$. $\mathfrak{g}$ is commutative if $[X,Y]=0$ for all $X,Y\in\mathfrak{g}$.

Example 3.2. *Let* $\mathfrak{g}=\mathbb{R}^3$*, let* $[\cdot,\cdot]:\mathbb{R}^3\times\mathbb{R}^3\to\mathbb{R}^3$ *be given by*

$$[x,y]=x\times y$$

where $x\times y$ *is the cross product. Then* $\mathfrak{g}$ *is a Lie algebra.*

Example 3.3. *Let* $\mathcal{A}$ *be an associative algebra and let* $\mathfrak{g}$ *be a subspace of* $\mathcal{A}$ *such that* $XY-YX\in\mathfrak{g}$ *for all* $X,Y\in\mathfrak{g}$*. Then* $\mathfrak{g}$ *is a Lie algebra with bracket operation given by* $[X,Y]=XY-YX$

Example 3.4. *Let* $\mathfrak{sl}(n,\mathbb{C})$ *denote the space of all* $X\in M_n(\mathbb{C})$ *for which* $tr(X)=0$*. Then* $\mathfrak{sl}(n,\mathbb{C})$ *is a Lie algebra with bracket* $[X,Y]=XY-YX$*.*

Definition 3.5. *A **subalgebra** of a real or complex Lie algebra* $\mathfrak{g}$ *is a subspace* $\mathfrak{h}$ *of* $\mathfrak{g}$ *such that* $[H_1,H_2]\in\mathfrak{h}$ *for all* $H_1,H_2\in\mathfrak{h}$*. If* $\mathfrak{g}$ *is a complex Lie algebra and* $\mathfrak{h}$ *is a real subspace of* $\mathfrak{g}$ *which is closed under brackets, then* $\mathfrak{h}$ *is said to be a **real subalgebra** of* $\mathfrak{g}$*.*

A subalgebra $\mathfrak{h}$ of a Lie algebra $\mathfrak{g}$ is said to be an **ideal** in $\mathfrak{g}$ if $[X,H]\in\mathfrak{h}$ for all $X\in\mathfrak{g}$ and $H\in\mathfrak{h}$.

The **center** of a Lie algebra $\mathfrak{g}$ is

$$\mathfrak{z}(\mathfrak{g})=\{X\in\mathfrak{g}:[X,Y]=0\ \forall Y\in\mathfrak{g}\}$$

Example. *The vector space* $\mathfrak{gl}(n,\mathbb{C})$ *of all* $n\times n$ *complex matrices is a Lie algebra with Lie bracket* $[X,Y]=XY-YX$*. The subspace* $\mathfrak{sl}(n,\mathbb{C})$ *of all* $n\times n$ *complex matrices with trace zero is a Lie subalgebra of* $\mathfrak{gl}(n,\mathbb{C})$*.*

Example. *Every Lie algebra* $\mathfrak{g}$ *has at least two ideals:* $\{0\}$ *and* $\mathfrak{g}$*.*
$\mathfrak{sl}(n,\mathbb{C})$ *is a non-trivial ideal of* $\mathfrak{gl}(n,\mathbb{C})$ *for* $n>1$*.*
For a Lie algebra $\mathfrak{g}$*, its center* $\mathfrak{z}(\mathfrak{g})$ *and its derived algebra* $\mathcal{D}(\mathfrak{g})=[\mathfrak{g},\mathfrak{g}]$ *are ideals.*

Example. *The center of the Heisenberg Lie algebra is*

$$\mathfrak{z}(\mathfrak{h})=\left\{\begin{pmatrix}0&0&c\\0&0&0\\0&0&0\end{pmatrix}:c\in\mathbb{R}\right\}=\{cZ:c\in\mathbb{R}\}$$

Definition 3.6. *If $\mathfrak{g}$ and $\mathfrak{h}$ are Lie algebras, then a linear map $\phi : \mathfrak{g} \to \mathfrak{h}$ is called a **Lie algebra homomorphism** if $\phi([X,Y]) = [\phi(X), \phi(Y)]$ for all $X, Y \in \mathfrak{g}$. If, in addition, ϕ is one-to-one and onto (i.e. bijective), then ϕ is called a **Lie algebra isomorphism**. A Lie algebra isomorphism of a Lie algebra with itself is called a **Lie algebra automorphism**.*

Definition 3.7. *If $\mathfrak{g}$ is a Lie algebra and X is an element of $\mathfrak{g}$, define a linear map $ad_X : \mathfrak{g} \to \mathfrak{g}$ by*

$$ad_X(Y) = [X, Y].$$

*The map $X \mapsto ad_X$ is the **adjoint map** or **adjoint representation**.*

We can view ad (the map $X \mapsto ad_X$) as a linear map of $\mathfrak{g}$ into $End(\mathfrak{g})$, the space of linear operators on $\mathfrak{g}$.

Proposition 3.8. *If $\mathfrak{g}$ is a Lie algebra, then*

$$ad_{[X,Y]} = ad_X ad_Y - ad_Y ad_X = [ad_X, ad_Y];$$

that is, $ad : \mathfrak{g} \mapsto End(\mathfrak{g})$ is a Lie algebra homomorphism.

Example. *If $\mathfrak{g}$ is a Lie algebra, then the identity map $x \mapsto x$ is a Lie algebra isomorphism, and also a Lie algebra automorphism.*

Let K be a field and $\mathfrak{gl}(n, K)$ be the Lie algebra given by all $n \times n$ matrices over K, with Lie bracket given by $[X,Y] = XY - YX$. For $X \in \mathfrak{gl}(K)$ and $g \in GL(n, K)$, the map $X \mapsto gXg^{-1}$ is an isomorphism from $\mathfrak{gl}(n, K)$ to itself, i.e. an automorphism of $\mathfrak{gl}(n, K)$.

Definition 3.9. *If $\mathfrak{g}_1$ and $\mathfrak{g}_2$ are Lie algebras, the **direct sum** of $\mathfrak{g}_1$ and $\mathfrak{g}_2$ is the vector space direct sum of $\mathfrak{g}_1$ and $\mathfrak{g}_2$, with bracket given by*

$$[(X_1, X_2), (Y_1, Y_2)] = ([X_1, Y_1], [X_2, Y_2]).$$

*If $\mathfrak{g}$ is a Lie algebra and $\mathfrak{g}_1$ and $\mathfrak{g}_2$ are subalgebras, we say that $\mathfrak{g}$ **decomposes as the Lie algebra direct sum** of $\mathfrak{g}_1$ and $\mathfrak{g}_2$ if $\mathfrak{g}$ is the direct sum of $\mathfrak{g}_1$ and $\mathfrak{g}_2$ as vector spaces and $[X_1, X_2] = 0$ for all $X_1 \in \mathfrak{g}_1$ and $X_2 \in \mathfrak{g}_2$.*

Example. *Examples of Lie algebra direct sums*

- $\mathfrak{g} = \mathbb{R} \oplus \mathbb{R}$
- $\mathfrak{gl}(2, \mathbb{C}) = \mathfrak{sl}(2, \mathbb{C}) \oplus \mathbb{C}I$

Definition 3.10. *Let $\mathfrak{g}$ be a finite-dimensional real or complex Lie algebra, and let $X_1, \ldots, X_N$ be a basis for $\mathfrak{g}$ (as a vector space). Then the unique constants c_{jkl} such that*

$$[X_j, X_k] = \sum_{l=1}^{N} c_{jkl} X_l$$

*are called the **structure constants** of $\mathfrak{g}$ (with respect to the chosen basis).*

The structure constants satisfy the following two conditions:

$$c_{jkl} + c_{kjl} = 0$$

$$\sum_n (c_{jkn} c_{nlm} + c_{kln} c_{njm} + c_{ljn} c_{nkm}) = 0$$

for all j, k, l, m.

Example. *A basis for* $\mathfrak{sl}(2,\mathbb{C})$ *is*

$$X_1 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}; \quad X_2 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}; \quad X_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

and we have the relations

$$[X_3, X_1] = 2X_1; \quad [X_3, X_2] = -2X_2; \quad [X_1, X_2] = X_3$$

The structure constants are

$$c_{311} = 2, \quad c_{322} = -2, \quad c_{123} = 1$$

with all others following from skew symmetry or equal to 0.

3.2 Simple, Solvable, and Nilpotent Lie Algebras

Definition 3.11. *A Lie algebra* $\mathfrak{g}$ *is called* ***irreducible*** *if the only ideals in* $\mathfrak{g}$ *are* $\mathfrak{g}$ *and* $\{0\}$*. A Lie algebra* $\mathfrak{g}$ *is called* ***simple*** *if it is irreducible and* $\dim \mathfrak{g} \geq 2$

Example. *Irreducible but not simple Lie algebras include the one-dimensional Lie algebras. For instance, take* $\mathfrak{g} = \mathbb{R}$ *with bracket* $[x, y] = 0$*.*

Proposition 3.12. *The Lie algebra* $\mathfrak{sl}(2,\mathbb{C})$ *is simple.*

Definition 3.13. *If* $\mathfrak{g}$ *is a Lie algebra, then the* ***commutator ideal*** *in* $\mathfrak{g}$*, denoted* $[\mathfrak{g}, \mathfrak{g}]$*, is the space of linear combinations of commutators, that is, the space of elements* Z *in* $\mathfrak{g}$ *that can be expressed as*

$$Z = c_1[X_1, Y_1] + \cdots + c_m[X_m, Y_m]$$

for some constants c_j *and vectors* $X_j, Y_j \in \mathfrak{g}$*.*

Example. *For* $\mathfrak{g} = \mathfrak{gl}(n,\mathbb{C})$*, the commutator ideal is* $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{sl}(n,\mathbb{C})$*.*

Definition 3.14. *For any Lie algebra* $\mathfrak{g}$*, we define a sequence of subalgebras* $\mathfrak{g}_0, \mathfrak{g}_1, \mathfrak{g}_2, \ldots$ *of* $\mathfrak{g}$ *inductively as follows:* $\mathfrak{g}_0 = \mathfrak{g}$*,* $\mathfrak{g}_1 = [\mathfrak{g}_0, \mathfrak{g}_0]$*,* $\mathfrak{g}_2 = [\mathfrak{g}_1, \mathfrak{g}_1]$*, etc. These subalgebras are called the* ***derived series*** *of* $\mathfrak{g}$*. A Lie algebra* $\mathfrak{g}$ *is called* ***solvable*** *if* $\mathfrak{g}_j = \{0\}$ *for some* j*.*

It can be shown, using the Jacobi identity and induction on j, that each $\mathfrak{g}_j$ is an ideal in $\mathfrak{g}$.

Definition 3.15. *For any Lie algebra* $\mathfrak{g}$*, we define a sequence of ideals* $\mathfrak{g}^j$ *in* $\mathfrak{g}$ *inductively as follows. We set* $\mathfrak{g}^0 = \mathfrak{g}$ *and then define* $\mathfrak{g}^{j+1}$ *to be the space of linear combinations of commutators of the form* $[X, Y]$ *with* $X \in \mathfrak{g}$ *and* $Y \in \mathfrak{g}^j$*. These algebras are called the* ***upper central series*** *of* $\mathfrak{g}$*. A Lie algebra* $\mathfrak{g}$ *is said to be* ***nilpotent*** *if* $\mathfrak{g}^j = \{0\}$ *for some* j*.*

Proposition 3.16. *If* $\mathfrak{g} \subset M_3(\mathbb{R})$ *denotes the space of* 3×3 *upper triangular matrices with zeros on the diagonal, then* $\mathfrak{g}$ *satisfies the assumptions of Example 3.3. The Lie algebra* $\mathfrak{g}$ *is a nilpotent Lie algebra.*

Proposition 3.17. *If* $\mathfrak{g} \subset M_2(\mathbb{C})$ *denotes the space of* 2×2 *matrices of the form*

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$$

with $a, b, c \in \mathbb{C}$*, then* $\mathfrak{g}$ *satisfies the assumptions of Example 3.3. The Lie algebra* $\mathfrak{g}$ *is solvable but not nilpotent.*

I will use the next two examples to develop Proposition 3.16 and 3.17.

Example. *Take*

$$\mathfrak{g} = \left\{ \begin{pmatrix} 0 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$$

The upper central series is as follows. Set $\mathfrak{g}^0 = \mathfrak{g}$. *Then*

$$\mathfrak{g}^1 = [\mathfrak{g}^0, \mathfrak{g}^0] = \left\{ \begin{pmatrix} 0 & 0 & b \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} : b \in \mathbb{R} \right\}$$

$$\mathfrak{g}^2 = \{0\}$$

Hence $\mathfrak{g}$ *is nilpotent.*

Example. *Take*

$$\mathfrak{g} = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} : a, b, c \in \mathbb{C} \right\}$$

which is actually a subalgebra of $\mathfrak{gl}(2, \mathbb{C})$. *The derived series is as follows. Define* $\mathfrak{g}_0 = \mathfrak{g}$, *then*

$$\mathfrak{g}_1 = [\mathfrak{g}_0, \mathfrak{g}_0] = \left\{ \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix} : b \in \mathbb{C} \right\}$$

$$\mathfrak{g}_2 = [\mathfrak{g}_1, \mathfrak{g}_1] = \{0\}$$

Hence $\mathfrak{g}$ *is solvable.*

3.3 The Lie Algebra of a Matrix Lie Group

3.4 Examples

3.5 Lie Group and Lie Algebra Homomorphisms

3.6 The Complexification of a Real Lie Algebra

3.7 The Exponential Map

3.8 Consequences of Theorem 3.42

3.9 Exercises

Exercise 3.8 Show that two isomorphic matrix Lie groups have isomorphic Lie algebras.

Proof. Let G and H be two isomorphic matrix Lie groups ($G \cong H$) with corresponding Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively. Since $G \cong H$, let $\Phi : G \to H$ be a Lie group isomorphism. By Theorem 3.28, there exists a unique real-linear map $\phi : \mathfrak{g} \to \mathfrak{h}$ such that

$$\Phi(e^X) = e^{\phi(X)}$$

for all $X \in \mathfrak{g}$. The map ϕ has the properties

1. $\phi(AXA^{-1}) = \Phi(A)\phi(X)\Phi(A)^{-1}$ for all $X \in \mathfrak{g}, A \in G$
2. $\phi([X, Y]) = [\phi(X), \phi(Y)]$, for all $X, Y \in \mathfrak{g}$
3. $\phi(X) = \frac{d}{dt}\Phi(e^{tX})\big|_{t=0}$ for all $X \in \mathfrak{g}$.

From property 2, it is clear that ϕ is a Lie algebra homomorphism. It remains to show that ϕ is bijective.

Since Φ is a Lie group isomorphism, it must be bijective, hence injective. So $ker(\Phi) = \{e_G\}$. By Proposition 3.31, we have

$$ker(\phi) = Lie(ker(\Phi)) = Lie(\{e_G\}) = \{0\}$$

where 0 is the zero element of $\mathfrak{g}$. This implies ϕ is injective. I assume that we are only working with finite-dimensional Lie groups and finite-dimensional Lie algebras. In this case, the dimension of a Lie group equals the dimension of its Lie algebra. Also, since $G \cong H$, $\dim G = \dim H$. So we have

$$\dim \mathfrak{g} = \dim G = \dim H = \dim \mathfrak{h}$$

Since ϕ is injective and $\dim \mathfrak{g} = \dim \mathfrak{h}$, ϕ is bijective.

Hence ϕ is an isomorphism, and the Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$ are isomorphic. □

Remark. *Based on the discussion during the groupe de travail, to complete the proof, we still need to show that ϕ^{-1} is a Lie group homomorphism, i.e. it satisfies*

$$\phi^{-1}([X,Y]) = [\phi^{-1}(X), \phi^{-1}(Y)]$$

To do this, we can apply ϕ to both sides, and see that both yield $[X,Y]$. Since ϕ is injective, the condition has been proven. However, I am not sure why this extra step is needed.

Exercise 3.13 Let G be a matrix Lie group and let $\mathfrak{g}$ be its Lie algebra. For each $A \in G$, show that Ad_A is a Lie algebra automorphism of $\mathfrak{g}$.

Proof. For each $A \in G$,

$$\begin{aligned}
[Ad_A(X), Ad_A(Y)] &= Ad_A(X)Ad_A(Y) - Ad_A(Y)Ad_A(X) \\
&= (AXA^{-1})(AYA^{-1}) - (AYA^{-1})(AXA^{-1}) \\
&= A(XY)A^{-1} - A(YX)A^{-1} \\
&= A(XY - YX)A^{-1} \\
&= A[X,Y]A^{-1} \\
&= Ad_A([X,Y])
\end{aligned}$$

for all $X, Y \in \mathfrak{g}$. Hence $Ad_A : \mathfrak{g} \to \mathfrak{g}$ is a Lie algebra homomorphism. It remains to prove that Ad_A is bijective.

Let $X, Y \in \mathfrak{g}$, with $X \neq Y$. We have $Ad_A(X) = AXA^{-1}$ and $Ad_A(Y) = AYA^{-1}$. Suppose $Ad_A(X) = Ad_A(Y)$, then $AXA^{-1} = AYA^{-1}$, then by left and right multiplication by A, $X = Y$, which is a contradiction. So Ad_A is an injective mapping.

Let $X \in \mathfrak{g}$. Define $P = A^{-1}XA$. By Theorem 3.20, $AXA^{-1} \in \mathfrak{g}$ for all $A \in G$. Since G is a group, $A^{-1} \in G$, so we see that $P = A^{-1}XA \in \mathfrak{g}$ for each $A \in G$. It is clear that

$$Ad_A(P) = APA^{-1} = X$$

This means $\forall X \in \mathfrak{g}$, $\exists P \in \mathfrak{g}$ such that $Ad_A(P) = X$. So Ad_A is a surjective mapping.

Hence $Ad_A : \mathfrak{g} \to \mathfrak{g}$ is a Lie algebra isomorphism, and furthermore a Lie algebra automorphism. □

4 Basic Representation Theory

4.1 Representations

Definition 4.1. *Let G be a matrix Lie group. A **finite-dimensional complex representation** of G is a Lie group homomorphism*

$$\Pi : G \to GL(V)$$

*where V is a finite-dimensional complex vector space (with $\dim(V) \geq 1$). A **finite-dimensional real representation** of G is a Lie group homomorphism Π of G into $GL(V)$ with V is a finite-dimensional real vector space.*

If $\mathfrak{g}$ is a real or complex Lie algebra, then a **finite-dimensional complex representation** of $\mathfrak{g}$ is a Lie algebra homomorphism $\pi : \mathfrak{g} \to gl(V)$, where V is a finite-dimensional complex vector space. If $\mathfrak{g}$ is a real Lie algebra, then a **finite-dimensional real representation** of $\mathfrak{g}$ is a Lie algebra homomorphism $\pi : \mathfrak{g} \to gl(V)$, where V is a finite-dimensional real vector space.

If Π or π is an injective homomorphism, the representation is called **faithful**.

Definition 4.2. *Let Π be a finite-dimensional real or complex representation of a matrix Lie group G, acting on a space V. A subspace W of V is called **invariant** if $\Pi(A)w \in W$ for all $w \in W$ and all $A \in G$. An invariant subspace W is called **nontrivial** if $W \neq \{0\}$ and $W \neq V$. A representation with no nontrivial invariant subspaces is called **irreducible**.*

The terms **invariant**, **nontrivial**, and **irreducible** are defined analogously for representations of Lie algebras.

We many consider complex representations of $\mathfrak{g}$.

Definition 4.3. *Let G be a matrix Lie group, let Π be a representation of G acting on the space V, and let Σ be a representation of G acting on the space W. A linear map $\phi : V \to W$ is called an **intertwining map** of representations if*

$$\phi(\Phi(A)v) = \Sigma(A)\phi(v)$$

for all $A \in G$ and all $v \in V$. The analogous property defines intertwining maps of representations of a Lie algebra.

The following proposition is a result of Theorem 3.28.

Proposition 4.4. *Let G be a matrix Lie group with Lie algebra $\mathfrak{g}$ and let Π be a finite-dimensional real or complex representation of G, acting on the space V. Then there is a unique representation π of $\mathfrak{g}$ acting on the same space such that*

$$\Pi(e^X) = e^{\pi(X)}$$

for all $X \in \mathfrak{g}$. The representation π can be computed as

$$\pi(X) = \frac{d}{dt}\Pi(e^{tX})\Big|_{t=0}$$

and satisfies

$$\pi(AXA^{-1}) = \Pi(A)\pi(X)\Pi(A)^{-1}$$

for all $X \in \mathfrak{g}$ and all $A \in G$.

Proposition 4.5. *1. Let G be a connected matrix Lie group with Lie algebra $\mathfrak{g}$. Let Π be a representation of G and π the associated representation of $\mathfrak{g}$. Then Π is irreducible if and only if π is irreducible.*

2. *Let G be a connected matrix Lie group, let Π_1 and Π_2 be representations of G, and let π_2 and π_2 be the associated Lie algebra representations. Then π_1 and π_2 are isomorphic if and only if Π_1 and Π_2 are isomorphic.*

Proposition 4.6. *Let $\mathfrak{g}$ be a real Lie algebra and $\mathfrak{g}_{\mathbb{C}}$ its complexification. Then every finite-dimensional complex representation π of $\mathfrak{g}$ has a unique extension to a complex-linear representation of $\mathfrak{g}_{\mathbb{C}}$, also denoted π. Furthermore, π is irreducible as a representation of $\mathfrak{g}_{\mathbb{C}}$ if and only if it is irreducible as a representation of $\mathfrak{g}$.*

The extension of π to $\mathfrak{g}_{\mathbb{C}}$ is given by $\pi(X+iY) = \pi(X) + i\pi(Y)$ for all $X, Y \in \mathfrak{g}$.

Definition 4.7. *If V is a finite-dimensional inner product space and G is a matrix Lie group, a representation $\Pi : G \to GL(V)$ is **unitary** if $\Pi(A)$ is a unitary operator on V for every $A \in G$.*

Proposition 4.8. *Suppose G is a matrix Lie group with Lie algebra $\mathfrak{g}$. Suppose V is a finite-dimensional inner product space, Π is a representation of G acting on V, and π is the associated representation of $\mathfrak{g}$. If Π is unitary, then $\pi(X)$ is skew self-adjoint for all $X \in \mathfrak{g}$. Conversely, if G is connected and $\pi(X)$ is skew self-adjoint for all $X \in \mathfrak{g}$, then Π is unitary.*

A representation π of a real Lie algebra $\mathfrak{g}$ acting on a finite-dimensional inner product space is **unitary** is $\pi(X)$ is skew self-adjoint for all $X \in \mathfrak{g}$.

4.2 Examples of Representations

A matrix Lie group G is a subset of some $GL(n, \mathbb{C})$. The inclusion map of G into $GL(n, \mathbb{C})$, $\Pi(A) = A$, is called the **standard representation** of G. Similarly, if $\mathfrak{g} \subset M_n(\mathbb{C})$ is a Lie algebra of matrices, the map $\pi(X) = X$ is called the **standard representation** of $\mathfrak{g}$.

For any matrix Lie group G, we can define the **trivial representation** $\Pi : G \to GL(1, \mathbb{C})$ by

$$\Pi(A) = I$$

for all $A \in G$. This is an irreducible representation. For a Lie algebra $\mathfrak{g}$, we can define the **trivial representation** of $\mathfrak{g}$, $\pi : \mathfrak{g} \to gl(1, \mathbb{C})$ by

$$\pi(X) = 0$$

for all $X \in \mathfrak{g}$. This is an irreducible representation.

Definition 4.9. *If G is a matrix Lie group with Lie algebra $\mathfrak{g}$, the **adjoint representation** of G is the map $Ad : G \to GL(\mathfrak{g})$ given by $A \mapsto Ad_A$. Similarly, the **adjoint representation** of a finite-dimensional Lie algebra $\mathfrak{g}$ is the map $ad : \mathfrak{g} \to gl(\mathfrak{g})$ given by $X \mapsto ad_X$.*

Example 4.10. *Let V_m denote the space of homogeneous polynomials of degree m in two complex variables. For each $U \in SU(2)$, define a linear transformation $\Pi_m(U)$ on the space V_m by the formula*

$$[\Pi_m(U)f](z) = f(U^{-1}z), \quad z \in \mathbb{C}^2$$

Then Π_m is a representation of $SU(2)$.

Proposition 4.11. *For each $m \geq 0$, the representation π_m is irreducible.*

4.3 New Representations from Old

4.3.1 Direct Sums

Definition 4.12. *Let G be a matrix Lie group and let $\Pi_1, \Pi_2, \ldots, \Pi_m$ be representations of G acting on vector spaces $V_1, V_2, \ldots, V_m$. Then the* ***direct sum*** *of $\Pi_1, \Pi_2, \ldots, \Pi_m$ is a representation $\Pi_1 \oplus \cdots \oplus \Pi_m$ of G acting on the space $V_1 \oplus \cdots \oplus V_m$, defined by*

$$[\Pi_1 \oplus \cdots \oplus \Pi_m(A)](v_1, \ldots, v_m) = (\Pi_1(A)v_1, \ldots, \Pi_m(A)v_m)$$

for all $A \in G$.

Similarly, if $\mathfrak{g}$ is a Lie algebra, and $\pi_1, \pi_2, \ldots, \pi_m$ are representations of $\mathfrak{g}$ acting on vector spaces $V_1, V_2, \ldots, V_m$, then we define the **direct sum** of $\pi_1, \pi_2, \ldots, \pi_m$, acting on $V_1 \oplus \cdots \oplus V_m$ by

$$[\pi_1 \oplus \cdots \oplus \pi_m(X)](v_1, \ldots, v_m) = (\pi_1(X)v_1, \ldots, \pi_m(X)v_m)$$

for all $X \in \mathfrak{g}$.

4.3.2 Tensor Products

The space $U \otimes V$ is the space of linear combinations of products $u \otimes v$, that is, the space of elements of the form

$$a_1 u_1 \otimes v_1 + a_2 u_2 \otimes v_2 + \cdots + a_n u_n \otimes v_n.$$

The operator $\otimes$ should also be bilinear:

$$(u_1 + a u_2) \otimes v = u_1 \otimes v + a u_2 \otimes v,$$

$$u \otimes (v_1 + a v_2) = u \otimes v_1 + a u \otimes v_2.$$

We do not assume that the product is commutative. So even if $U = V$ so that $U \otimes V$ and $V \otimes U$ are the same space, $u \otimes v$ will not equal $v \otimes u$ in general.

Definition 4.13. *If U and V are finite-dimensional real or complex vector spaces, then a* ***tensor product*** *of U with V is a vector space W, together with a bilinear map $\phi : U \times V \to W$ with the following property: if ψ is any bilinear map of $U \times V$ into a vector space X, there exists a unique linear map $\tilde{\psi}$ of W into X such that the following diagram commutes.*

$$\begin{tikzcd} U \times V \arrow[rr, "\phi"] \arrow[dr, "\psi"'] & & W \arrow[dl, "\tilde{\psi}"] \\ & X & \end{tikzcd}$$

Theorem 4.14. *If U and V are any finite-dimensional real or complex vector spaces, then a tensor product (W, ϕ) exists. Furthermore, (W, ϕ) is unique up to canonical isomorphism. Suppose that (W, ϕ) is a tensor product and that $e_1, e_2, \ldots, e_n$ is a basis for U and $f_1, f_2, \ldots, f_m$ is a basis for V. Then*

$$\{\phi(e_j, f_k) : 1 \leq j \leq n, 1 \leq k \leq m\}$$

is a basis for W. In particular, $\dim(U \otimes V) = (\dim U)(\dim V)$.

Proof. Let U and V be finite-dimensional real or complex vector spaces. Let $\{e_j\}_1^n$ be a basis for U, let $\{f_k\}_1^m$ be a basis for V.

Define $W = span\{w_{jk} : 1 \leq j \leq n, 1 \leq k \leq m\}$. Define a bilinear mapping $\phi : U \times V \to W$ as $\phi(e_j, f_k) = w_{jk}$.

By bilinearity, for $u = \sum_{j=1}^{n} \alpha_j e_j \in U$ and $v = \sum_{k=1}^{m} \beta_k f_k \in V$, we have

$$\begin{aligned}\phi(u,v) &= \sum_{j=1}^{n}\sum_{k=1}^{m} \alpha_j \beta_k \phi(e_j, f_k)\\ &= \sum_{j=1}^{n}\sum_{k=1}^{m} \alpha_j \beta_k w_{jk}\end{aligned}$$

Let $\psi : U \times V \to X$ be any bilinear map, let X be a vector space. Does there exist a unique linear map $\tilde{\psi} : W \to X$ such that the diagram (for the universal property) commutes? Suppose $X = span\{b_1, \ldots, b_N\}$. Define $\tilde{\psi} : W \to X$ by $\tilde{\psi}(w_{jk}) = \psi(e_j, f_k)$. Then,

$$\begin{aligned}(\tilde{\psi} \circ \phi)(u,v) &= \tilde{\psi}(\phi(u,v))\\ &= \tilde{\psi}(\sum_{j=1}^{n}\sum_{k=1}^{m} \alpha_j \beta_k w_{jk})\\ &= \sum_{j=1}^{n}\sum_{k=1}^{m} \alpha_j \beta_k \tilde{\psi}(w_{jk})\\ &= \sum_{j=1}^{n}\sum_{k=1}^{m} \alpha_j \beta_k \psi(e_j, f_k)\\ &= \psi(\sum_{j=1}^{n} \alpha_j e_j, \sum_{k=1}^{m} \beta_k f_k)\\ &= \psi(u,v)\end{aligned}$$

Thus $\tilde{\psi} \circ \phi = \psi$, and thus the diagram commutes. This proves existence.

Assume there exists $\tilde{\psi}'$ such that the diagram commutes. That is,

$$\tilde{\psi}' \circ \phi = \psi$$

and so we have

$$(\tilde{\psi}' \circ \phi)(u,v) = \psi(u,v) = (\tilde{\psi} \circ \phi)(u,v)$$

This means

$$\tilde{\psi}'(w_{jk}) = \tilde{\psi}'(\phi(e_j, f_k)) = \tilde{\psi}(\phi(e_j, f_j)) = \tilde{\psi}(w_{jk})$$

$\{w_{jk}\}$ is a basis, and $\tilde{\psi}$, $\tilde{\psi}'$ are linear maps, hence $\tilde{\psi}' = \tilde{\psi}$.

Hence there exists a unique linear map $\tilde{\psi} : W \to X$ such that the diagram commutes. By Definition 4.13 (the universal property), the tensor product (W, ϕ) exists.

Now we prove the tensor product is unique. Suppose (W_1, ϕ_1), (W_2, ϕ_2) are two tensor products. Consider again the universal property diagram.

If $X = W_2$, then $\tilde{\psi}_2 = Id$ and

$$\tilde{\psi}_1 \circ \phi_1 = \phi_2$$

If $X = W_1$, then $\tilde{\psi}_1 = Id$ and

$$\tilde{\psi}_2 \circ \psi_2 = \phi_1$$

So

$$\phi_1 = \tilde{\psi}_2 \circ \phi_2 = \tilde{\psi}_2 \circ (\tilde{\psi}_1 \circ \phi_1)$$

and so

$$(\tilde{\psi}_2 \circ \tilde{\psi}_1) \circ \phi_1 = \phi_1$$

On the other hand,

$$Id \circ \phi_1 = \phi_2$$

Since both $\tilde{\psi}_1$ and $\tilde{\psi}_2$ are unique, $\tilde{\psi}_2 \circ \tilde{\psi}_1$ must be unique, and must equal Id. So $\tilde{\psi}_2 = \tilde{\psi}_1^{-1}$. This shows that $\Phi = \tilde{\psi}_1 : W_1 \to W_2$ is an invertible linear map, hence bijective, hence an isomorphism.

Hence (W, ϕ) is unique up to canonical isomorphism.

We already know that

$$W = span\{w_{jk} : 1 \leq j \leq n, 1 \leq k \leq m\}$$

But $w_{jk} = \phi(e_j, f_k)$, so

$$\{\phi(e_j, f_k) : 1 \leq j \leq n, 1 \leq k \leq m\}$$

is a basis for W.

Finally, this basis has nm elements, therefore

$$\dim(U \otimes V) = nm = \dim(U)\dim(V)$$

□

Notation 4.15. *Since the tensor product of U and V is essentially unique, we will let $U \otimes V$ denote an arbitrary tensor product space and we will write $u \otimes v$ instead of $\phi(u, v)$. In this notation, Theorem 4.14 says that*

$$\{e_j \otimes f_k : 1 \leq j \leq n, 1 \leq k \leq m\}$$

is a basis for $U \otimes V$, as expected.

The defining property of $U \otimes V$ is called the **universal property** of tensor products.

Proposition 4.16. *Let U and V be finite-dimensional real or complex vector spaces. Let $A : U \to U$ and $B : V \to V$ be linear operators. Then there exists a unique linear operator from $U \otimes V$ to $U \otimes V$, denoted $A \otimes B$, such that*

$$(A \otimes B)(u \otimes v) = (Au) \otimes (Bv)$$

for all $u \in U$ and $v \in V$. If A_1 and A_2 are linear operators on U, and B_1 and B_2 are linear operators on V, then

$$(A_1 \otimes B_1)(A_2 \otimes B_2) = (A_1A_2) \otimes (B_1B_2).$$

Proof. Define $\psi : U \times V \to U \otimes V$, with $\psi(u, v) = (Au) \otimes (Bv)$. A and B are linear, while $\otimes$ is bilinear, so ψ is bilinear.

By the universal property, there exists a unique linear map $\tilde{\psi} : U \otimes V \to U \otimes V$ such that

$$\tilde{\psi}(u \otimes v) = \psi(u, v) = (Au) \otimes (Bv)$$

Thus $\tilde{\psi}$ is the desired map $A \otimes B$. To check the required formula, we first check the left hand side,

$$\begin{aligned}(A_1 \otimes B_1)(A_2 \otimes B_2)(u \otimes v) &= (A_1 \otimes B_1)((A_2u) \otimes (B_2v)) \\ &= (A_1A_2u) \otimes (B_1B_2v) \\ &= ((A_1A_2)u) \otimes ((B_1B_2)v)\end{aligned}$$

On the other hand,

$$((A_1A_2) \otimes (B_1B_2))(u \otimes v) = ((A_1A_2)u) \otimes ((B_1B_2)v)$$

So,

$$(A_1 \otimes B_1)(A_2 \otimes B_2)(u \otimes v) = ((A_1A_2) \otimes (B_1B_2))(u \otimes v)$$

By Theorem 4.14, elements of the form $u \otimes v$ span $U \otimes V$, hence

$$(A_1 \otimes B_1)(A_2 \otimes B_2) = (A_1A_2) \otimes (B_1B_2)$$

□

So far, we have been discussing tensor products of vector spaces. We then discuss tensor products of representations.

Definition 4.17. *Let G and H be matrix Lie groups. Let Π_1 be a representation of G acting on a space U and let Π_2 be a representation of H acting on a space V. Then the **tensor product** of Π_1 and Π_2 is a representation $\Pi_1 \otimes \Pi_2$ of $G \times H$ acting on $U \otimes V$ defined by*

$$(\Pi_1 \otimes \Pi_2)(A, B) = \Pi_1(A) \otimes \Pi_2(B)$$

for all $A \in G$ and $B \in H$.

Proposition 4.18. *Let G and H be matrix Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively. Let Π_1 and Π_2 be representations of G and H, respectively, and consider the representation $\Pi_1 \otimes \Pi_2$ of $G \times H$. If $\pi_1 \otimes \pi_2$ denotes the associated representation of $\mathfrak{g} \otimes \mathfrak{h}$, then*

$$(\pi_1 \otimes \pi_2)(X, Y) = \pi_1(X) \otimes I + I \otimes \pi_2(Y)$$

for all $X \in \mathfrak{g}$ and $Y \in \mathfrak{h}$.

Proof. Let $u(t)$ be a smooth curve in U, let $v(t)$ be a smooth curve in V. We will begin by following the same proof as for the product rule for scalar-valued functions.

$$\begin{aligned}
\frac{d}{dt}(u(t) \otimes v(t)) &= \lim_{h \to 0} \frac{u(t+h) \otimes v(t+h) - u(t) \otimes v(t)}{h} \\
&= \lim_{h \to 0} \frac{u(t+h) \otimes v(t+h) - u(t+h) \otimes v(t) + u(t+h) \otimes v(t) - u(t) \otimes v(t)}{h} \\
&= \lim_{h \to 0} \frac{u(t+h) \otimes (v(t+h) - v(t)) + (u(t+h) - u(t)) \otimes v(t)}{h} \\
&= \lim_{h \to 0} u(t+h) \otimes \frac{v(t+h) - v(t)}{h} + \lim_{h \to 0} \frac{u(t+h) - u(t)}{h} \otimes v(t) \\
&= u(t) \otimes \frac{dv}{dt} + \frac{du}{dt} \otimes v(t) \\
&= \frac{du}{dt} \otimes v(t) + u(t) \otimes \frac{dv}{dt}
\end{aligned}$$

We will use this result in the following computation.

$$\begin{aligned}
(\pi_1 \otimes \pi_2)(X, Y)(u \otimes v) &= \frac{d}{dt}(\Pi_1 \otimes \Pi_2)(e^{tX}, e^{tY})(u \otimes v)\Big|_{t=0} \\
&= \frac{d}{dt}\Pi_1(e^{tX})u \otimes \Pi_2(e^{tY})v\Big|_{t=0} \\
&= \frac{d}{dt}\Pi_1(e^{tX})u\Big|_{t=0} \otimes \Pi_2(e^{tY})v\Big|_{t=0} + \Pi_1(e^{tX})u\Big|_{t=0} \otimes \frac{d}{dt}\Pi_2(e^{tY})v\Big|_{t=0} \\
&= \frac{d}{dt}\Pi_1(e^{tX})u\Big|_{t=0} \otimes v + u \otimes \frac{d}{dt}\Pi_2(e^{tY})v\Big|_{t=0} \\
&= \pi_1(X)u \otimes v + u \otimes \pi_2(Y)v \\
&= (\pi_1(X) \otimes I)(u \otimes v) + (I \otimes \pi_2(Y))(u \otimes v) \\
&= (\pi_1(X) \otimes I + I \otimes \pi_2(Y))(u \otimes v)
\end{aligned}$$

By Theorem 4.14, elements of the form $u \otimes v$ span $U \otimes V$, hence

$$(\pi_1 \otimes \pi_2)(X, Y) = \pi_1(X) \otimes I + I \otimes \pi_2(Y)$$

$\square$

Definition 4.19. *Let $\mathfrak{g}$ and $\mathfrak{h}$ be Lie algebras and let π_1 and π_2 be representations of $\mathfrak{g}$ and $\mathfrak{h}$, acting on spaces U and V. Then the **tensor product** of π_1 and π_2, denoted $\pi_1 \otimes \pi_2$, is a representation of $\mathfrak{g} \otimes \mathfrak{h}$ acting on $U \otimes V$, given by*

$$(\pi_1 \otimes \pi_2)(X, Y) = \pi_1(X) \otimes I + I \otimes \pi_2(Y)$$

for all $X \in \mathfrak{g}$ and $Y \in \mathfrak{h}$.

Definition 4.20. *Let G be a matrix Lie group and let Π_1 and Π_2 be representations of G, acting on spaces V_1 and V_2. Then the **tensor product** representation of G, acting on $V_1 \otimes V_2$, is defined by*

$$(\Pi_1 \otimes \Pi_2)(A) = \Pi_1(A) \otimes \Pi_2(A)$$

for all $A \in G$. Similarly, if π_1 and π_2 are representations of a Lie algebra $\mathfrak{g}$, we define a tensor product representation of $\mathfrak{g}$ on $V_1 \otimes V_2$ by

$$(\pi_1 \otimes \pi_2)(X) = \pi_1(X) \otimes I + I \otimes \pi_2(X).$$

4.3.3 Dual Representations

In the following, for a finite-dimensional vector space V, denote V^* its dual space, the space of linear functionals on V. For a linear functional $A \in V^*$, A^{tr} denotes the dual or transpose operator, with the relation

$$(A^{tr}\phi)(v) = \phi(Av)$$

for $\phi \in V^*$, $v \in V$.

Definition 4.21. *Suppose G is a matrix Lie group and Π is a representation of G acting on a finite-dimensional vector space V. Then the **dual representation** Π^* to Π is the representation of G acting on V^* and given by*

$$\Pi^*(g) = [\Pi(g^{-1})]^{tr}$$

If π is a representation of a Lie algebra $\mathfrak{g}$ acting on a finite-dimensional vector space V, then π^* is the representation of $\mathfrak{g}$ acting on V^* and given by

$$\pi^*(X) = -\pi(X)^{tr}$$

In this definition, the inverse and the negative sign are essential. The dual representation is also called **contragredient representation**.

Proposition 4.22. *If Π is a representation of a matrix Lie group G, then*

1. *Π^* is irreducible if and only if Π is irreducible, and*
2. *$(\Pi^*)^*$ is isomorphic to Π.*

Similar statements apply to Lie algebra representations.

Proof. There is a one-to-one correspondence between subspaces of V and subspaces of V^* as follows: for any subspace W of V, the annihilator of W is the subspace of all ϕ in V^* such that ϕ is zero on W.

If W is an invariant subspace of V, then $\Pi(A)w \in W$ $\forall w \in W, A \in G$. $\forall \phi \in W^\perp, A \in G$,

$$\Pi^*(A)\phi(w) = [\Pi(A^{-1})]^{tr}\phi(w) = \phi(\Pi(A^{-1})(w)) = 0$$

where the last equality is because $\Pi(A^{-1})(w) \in W$. This means

$$\Pi^*(A)\phi \in W^\perp$$

Hence $W^\perp$ is an invariant subspace of V^*. Noting that $V^{**} \cong V$ and $W^{\perp\perp} = W$, we have W is an invariant subspace of V if and only if $W^\perp$ is an invariant subspace of V^*.

$$\begin{aligned}
\Pi \text{ is irreducible} &\iff \text{the only invariant subspaces of } V \text{ are } W = \{0\}, W = V \\
&\iff \text{the only invariant subspaces of } V^* \text{ are } W^\perp = V^*, W^\perp = \{0\} \\
&\iff \Pi^* \text{ is irreducible}
\end{aligned}$$

This proves the first point.

For the second point, first note that

$$\Pi : G \to GL(V)$$

$$(\Pi^*)^* : G \to GL(V^{**})$$

If we identify $V^{**} \cong V$, then we can easily prove equality. Otherwise, we need to prove there exists $\phi : V \to V^{**}$ an intertwining map of representations, i.e.

$$\phi(\Pi(A)v) = (\Pi^*)^*(A)\phi(v), \quad \forall A \in G, v \in V$$

that is also invertible.

Define $\varphi_v : V^* \to \mathbb{F}$ by $\varphi_v(f) = f(v)$, for $f \in V^*$. Define $\phi : V \to V^{**}, v \mapsto \varphi_v$. It can be shown that ϕ is linear, and is an isomorphism. This is the natural isomorphism between V and V^{**}.

Let $A \in G$, $v \in V$.

$$\begin{aligned}
\phi(\Pi(A)v)(f) &= \varphi_{\Pi(A)v}(f) \\
&= f(\Pi(A)v) \\
&= ([\Pi(A)]^{tr} f)(v) \\
&= \varphi_v([\Pi(A)]^{tr} f) \\
&= \phi(v)([\Pi(A)]^{tr} f) \\
&= \phi(v)(\Pi^*(A^{-1}) f) \\
&= ([\Pi^*(A^{-1})]^{tr} \phi(v))(f) \\
&= ((\Pi^*)^*(A)\phi(v))(f)
\end{aligned}$$

for all $f \in V^*$, and so

$$\phi(\Pi(A)v) = (\Pi^*)^*(A)\phi(v)$$

Hence ϕ is an intertwining map of representations, hence $(\Pi^*)^* \cong \Pi$. □

4.4 Complete Reducibility

4.5 Schur's Lemma

4.6 Representations of sl(2; C)

4.7 Group Versus Lie Algebra Representations

4.8 A Nonmatrix Lie Group

4.9 Exercises

Exercise 6 See proof of Proposition 4.22.
Exercise 7 See proof of Theorem 4.14.

References

[1] B.C. Hall. (2015). Lie Groups, Lie Algebras, and Representations An Elementary Introduction, Second Edition. Springer Cham.